

From Legal PDFs to Verifiable Answers

Evaluating answer quality, page-level grounding and latency across 100 legal questions

HYPOTHESIS Can a page-traceable Legal RAG pipeline balance answer quality, source grounding, validity and latency?

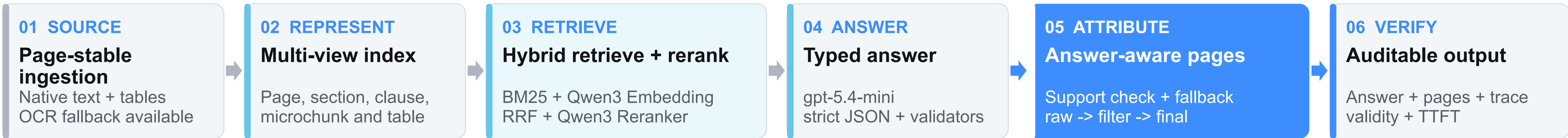
H2: Does answer-aware attribution sharpen citations while retaining at least one correct page per annotated question?

FINDING

H1 is partly supported: grounding missed its 0.85 threshold. H2 holds for precision, compactness and 95/95 hit coverage.

METHODOLOGY

100 questions | 30 publicly available legal PDFs | 590 pages | 95 with page annotations



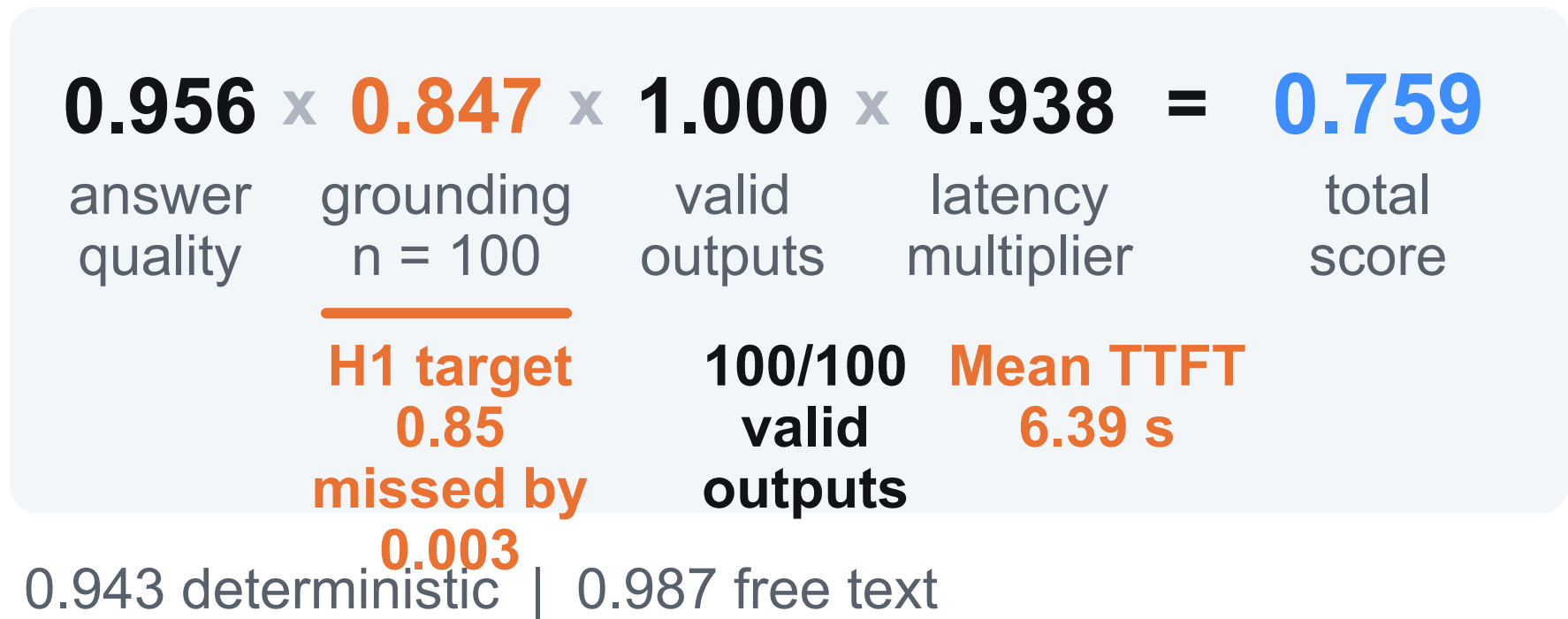
CONTINUOUS PROVENANCE | document_id + page_number

The same identifiers survive every stage, so a final citation can be traced back to the exact PDF page.

Evaluation A: end-to-end profile on 100 questions. Evaluation B: paired raw -> filter -> final page sets on the same 95 annotated questions.

RESULTS | SYSTEM

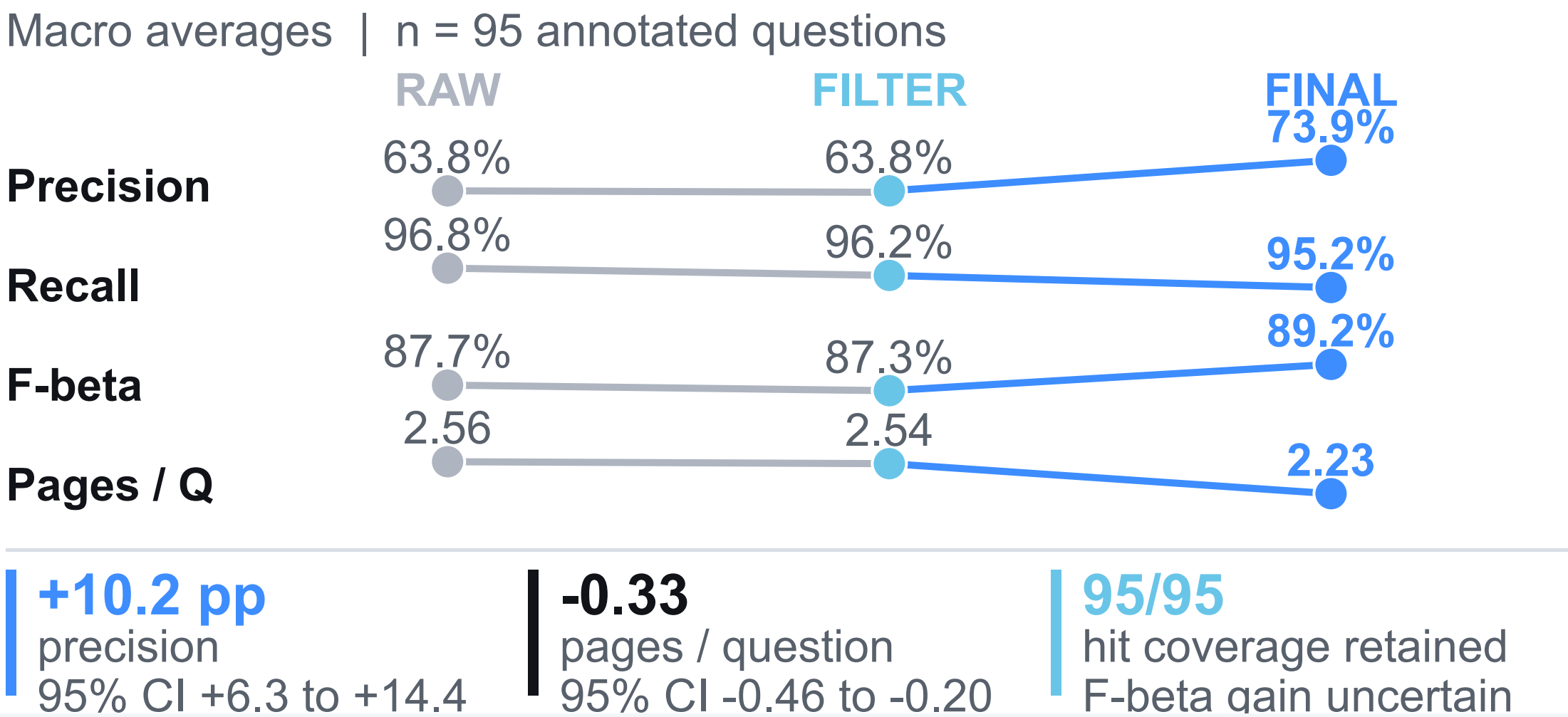
Grounding limits an otherwise strong system



H1 is partly supported: quality, validity and latency pass; benchmark grounding misses 0.85.

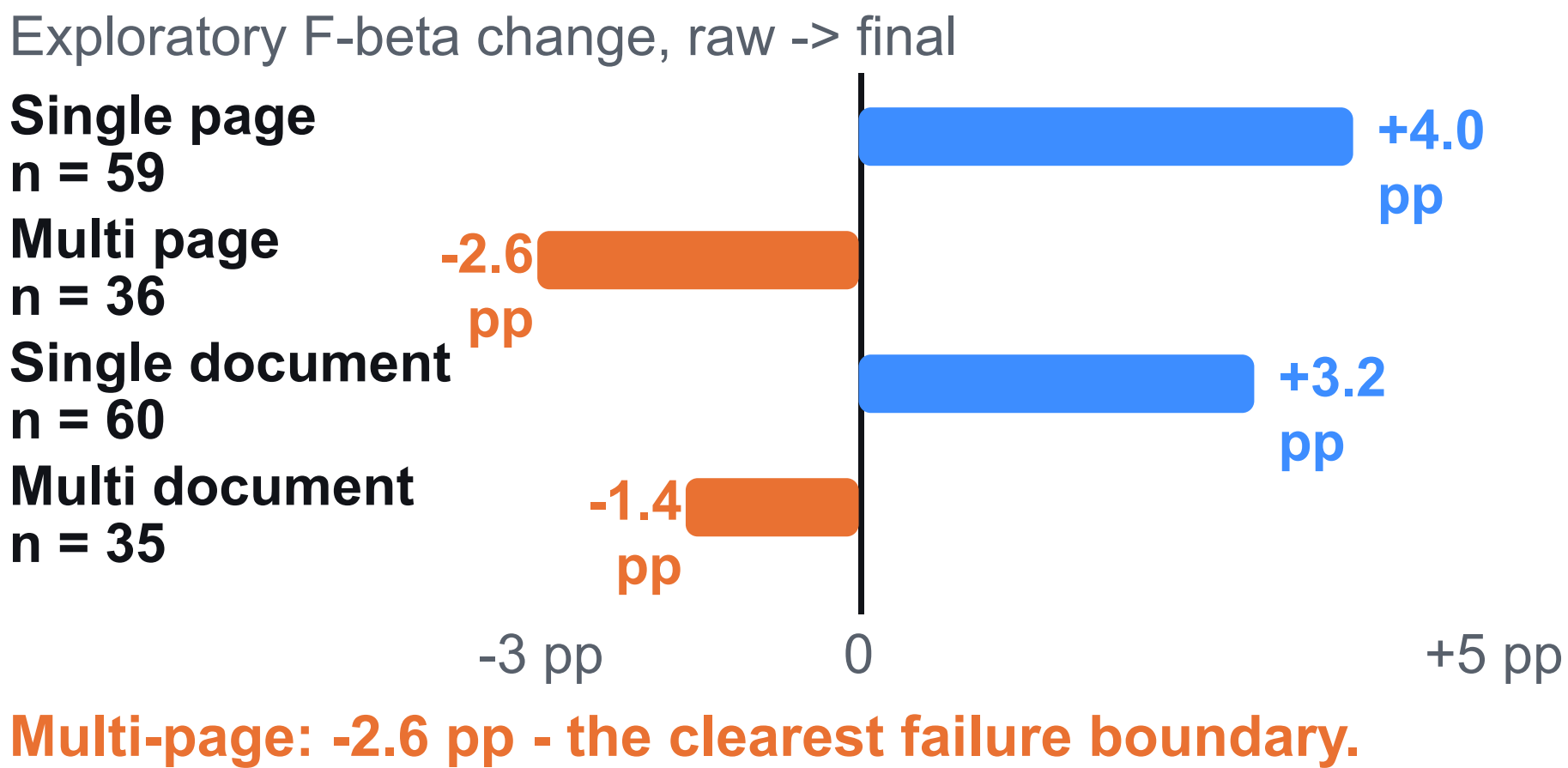
RESULTS | ATTRIBUTION

Answer-aware selection removes citation noise



RESULTS | BOUNDARY

Narrowing helps local - and hurts distributed evidence



Do not conflate: benchmark grounding 0.847 = all 100 answers; macro page F-beta 0.892 = final attribution on 95 annotated questions.

DECISION

Retrieve broadly. Attribute explicitly. Narrow conservatively when evidence spans pages or documents.

LIMITATIONS

ONE SAVED RUN

Retrospective thresholds and questions; no repeated seeds.

NO CAUSAL ABLATION

Pipeline stages were not independently compared against a baseline.

LIMITED GENERALITY

TTFT only; OCR not exercised; one DIFC corpus and one language setting.